

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15 21

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I harendha (1924521215) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

→ Cloud Auto Scaling and Load Balancing :

→ Auto-scaling automatically adds or removes servers based on website traffic.

→ Load balancing distributes user requests evenly across multiple servers.

→ This prevents server overload, reduces downtime, and improves website speed.

→ Users get a smooth shopping experience even during heavy traffic.

→ Monitoring tools check CPU, memory, and network usage and trigger auto-scaling when needed.

→ Example : During Amazon or Flipkart.

festivals sales, cloud services automatically add servers and balance traffic so

customers can shop without website crashes.

→ cloud storage for Backup data:-

→ cloud storage - stores backup data securely with multiple copies in different locations for high durability.

→ Data Redundancy :- ensures data is available even if the server or data center fails.

→ Encryption . protects backup data from unauthorized access during storage and transfer .

→ Access control : allows only authorized users to view or restore backup data .

→ cloud storage : is cost effective because there is no need to buy or maintain .

→ Ex :- If a company's office is affected by a flood or fire , it can restore all its data from cloud backups and continue operations quickly .

31 Content Delivery Networks (CDN's).

- A CDN stores copies of app content on multiple servers around the world.
- It reduces latency by delivering data from the nearest server to the user.
- Edge Servers. Improves user Experience by providing faster loading and lower delays in different regions.
- cloud providers use global data centers and automatic traffic routing to ensure high availability.
- This improves app performance, reliability and reduces server load.
- Ex:-
 - in games like PUBG mobile or Streaming apps like Netflix, CDN's and edge servers deliver content quickly, reducing buffering.

4) cloud security and compliance:-

→ cloud providers support regulatory compliance.

by following standards such as ISO 27001, PCI, PDS and GDPR. and providing audit reports.

→ logging and monitoring record all user activities and detect suspicious actions. for security and auditing.

→ Encryption protects sensitive financial data during storage and transmission.

→ Data sovereignty → is ensured by storing data in specific countries or regions. as required by law.

→ Ex :-

During an online bank transfer, Transaction details are encrypted, all activities.

5/ Containerization and Kubernetes:-

- Containers are lightweight and share the host operating system, while virtual machine (VM's) have their own OS.
- Containers use less memory and CPU, start faster, and provide better performance than VM's.
- Kubernetes automatically deploys, manages, scales and recovers containers if they fail.
- Containers are portable so the same app can run on AWS, Azure, Google Cloud.
- Ex:- An e-commerce application with separate microservices for user logic, product catalog, payment, and order.